

Culvert AI

Research portfolio for culvert discovery

A field-review system that combines geospatial machine learning, field evidence, and hydraulic review workflows for candidate culvert prioritization.

Demo objective

Give researchers and hydraulics engineers a clear basis for model critique, shared experiments, and a possible paper collaboration.

Geospatial ML + field validation

A working review workflow is in place

The prototype now connects data ingestion, candidate generation, model training, validation, and a browser-based field review queue.

Review app

Interactive map, ranked candidate list, ABU points, candidate detail panels, and field handoff links.

Prediction layer

Road-stream candidates, 81 engineered features, supervised probability estimates, and discovery scores.

Feedback loop

Approved, denied, no_culvert, ABU, and field observations can feed retraining and validation reports.

112,811

scored candidates

Full prediction layer used for model and coverage checks.

145

training points

Positive and negative evidence used by the current model.

81

engineered features

Geometry, density, terrain, context, and field support.

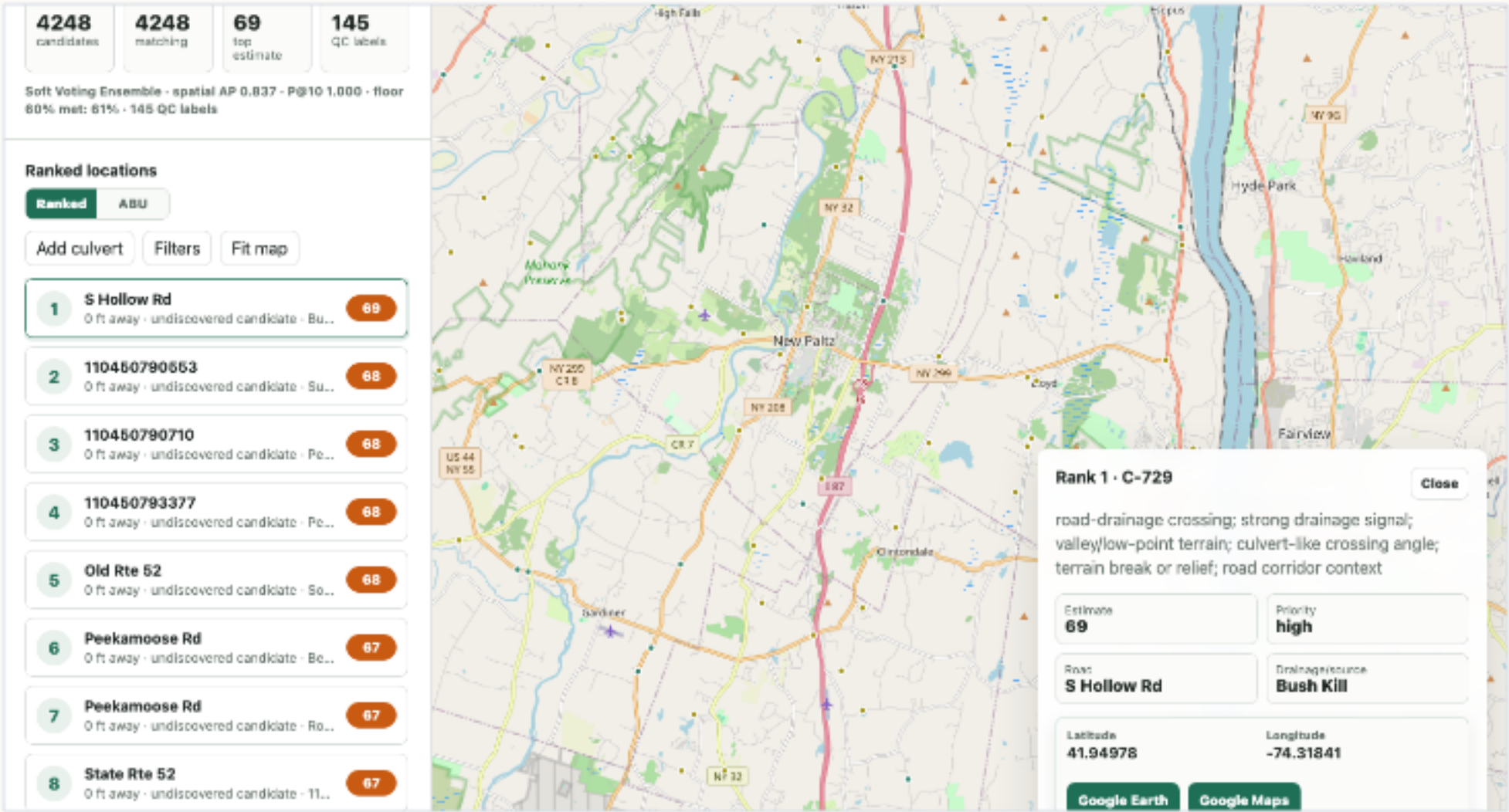
4,248

review targets

Decluttered export for practical browser review.

The interface supports field review

Reviewers can move from model rank to map context to field handoff without leaving the demo screen.

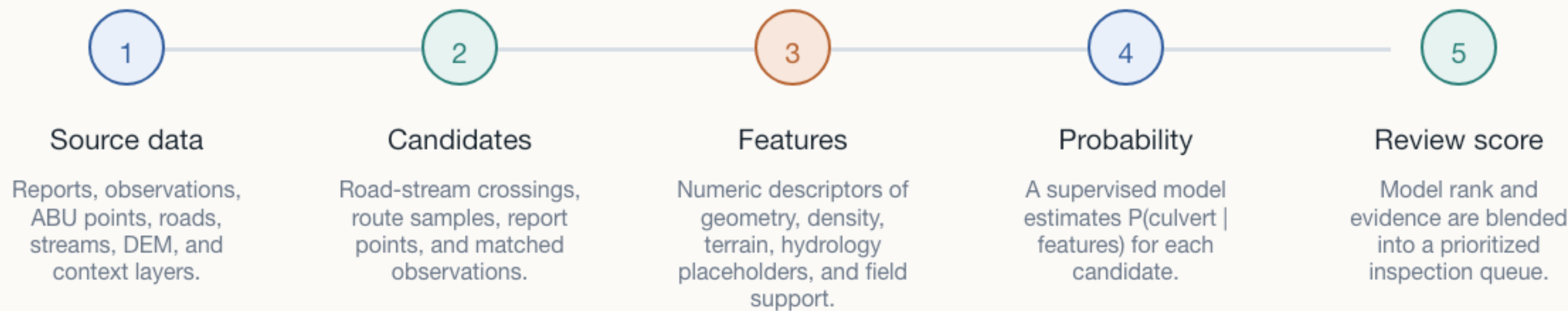


WHAT THE REVIEWER SEES

- **Ranked queue**
A sorted list of likely candidates, ABU points, filters, and map-fit controls.
- **Map context**
Roads, streams, terrain context, and candidate clusters remain visible while selecting targets.
- **Field handoff**
Each candidate exposes road, drainage/source, estimate, coordinates, and mapping links.

Prediction starts with spatial candidates

The system first proposes plausible locations, then learns which candidates resemble verified culvert evidence.



This architecture keeps the engineering question visible: the model ranks likely culvert locations, while field evidence and hydraulic review decide what becomes trusted inventory.

The learning task is supervised classification

The model is trained to distinguish likely culvert candidates from nonmatches using route-aware validation.

TRAINING SIGNAL

- **Positive examples**
Verified reports, confirmed ABU or user observations, and strict field matches.
- **Negative examples**
Nonmatches, no_culvert observations, denied points, and missed-prediction labels.
- **Spatial holdout**
Route-clustered validation reduces leakage from nearby points appearing in both train and test.

PREDICTION TARGET

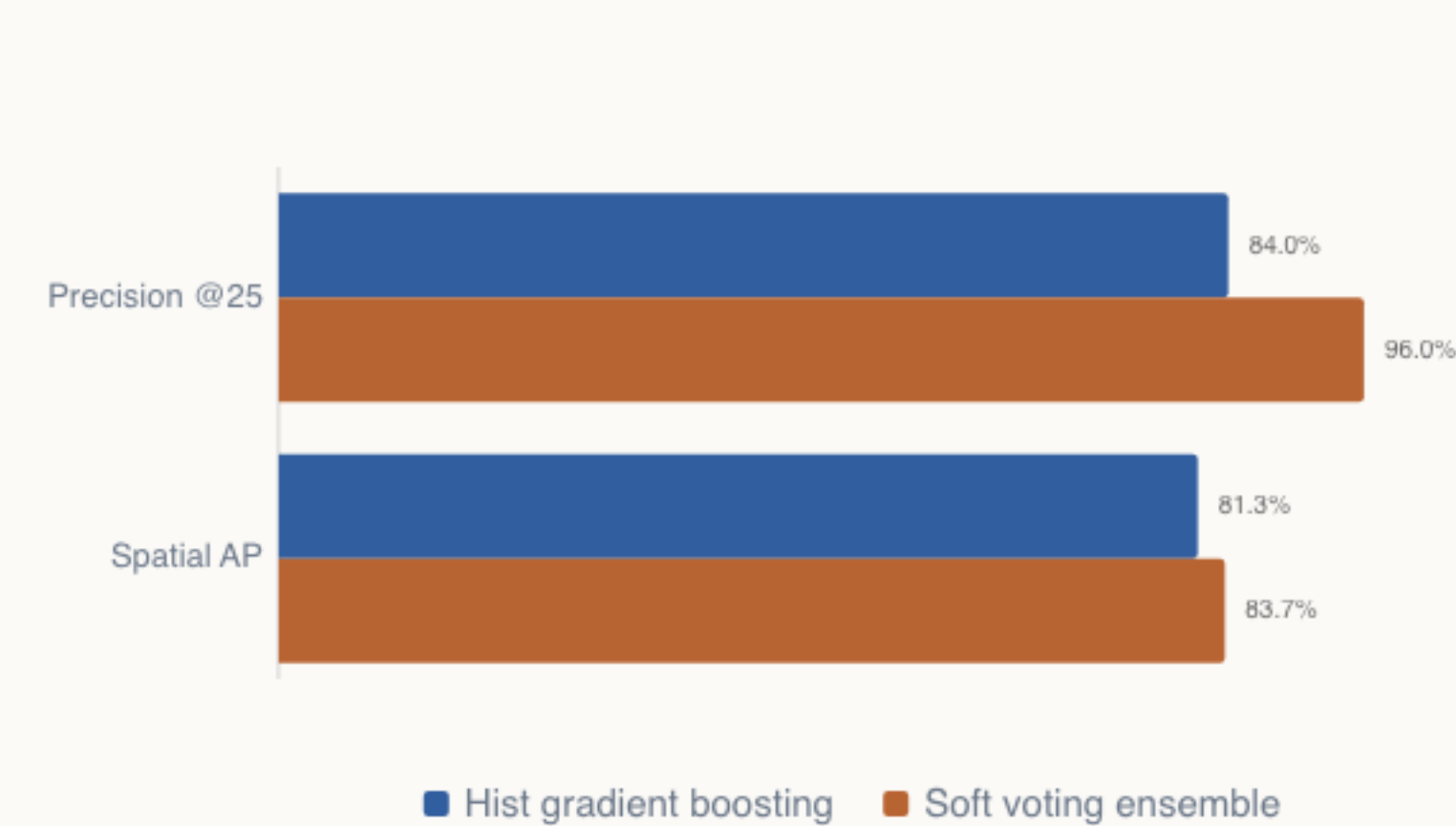
For every candidate point, the trained classifier returns a probability:

$$\text{culvert_probability} = P(\text{culvert} \mid \text{engineered features})$$

LLMs are not the primary predictor. They are useful for extraction and QA; the core prediction is geospatial ML.

Soft voting leads the current model set

The selected model performed best among tested candidates on spatial ranking metrics.



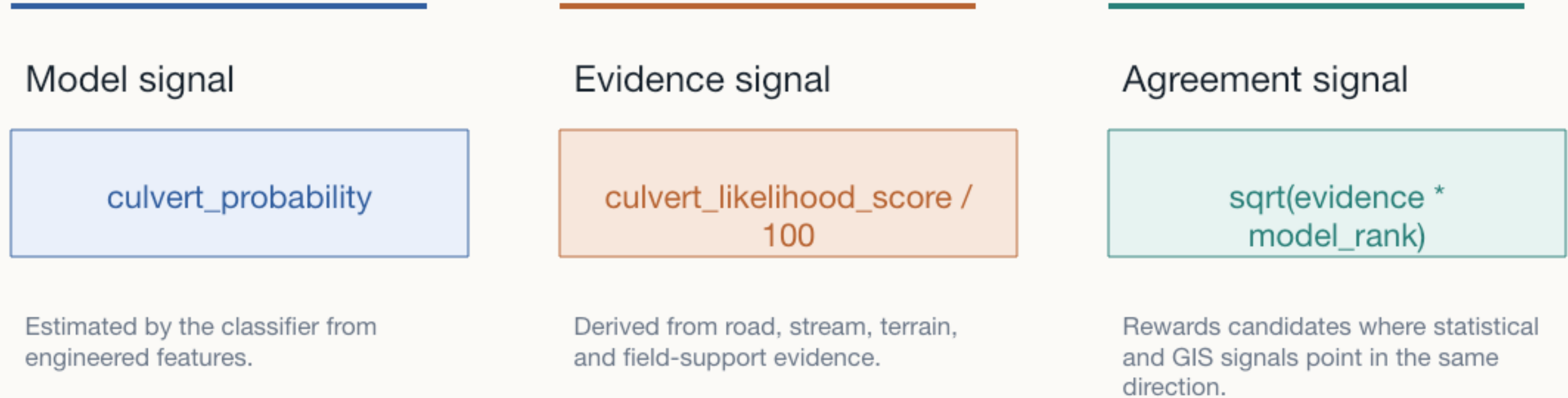
Why it was selected

Model selection favors spatial holdout average precision first, then cross-validation AP and F1. In the latest run, soft voting improved both spatial AP and early-rank precision while preserving perfect precision at the top 10.

Use this as the comparison baseline, not final proof of optimality.

The ranking blends model signal with evidence

The app separates model probability from the operational discovery score used to prioritize review.



$$\text{discovery_score} = 100 * (0.55 * \text{agreement} + 0.25 * \text{evidence} + 0.20 * \text{weighted_signal})$$

Current validation is promising but label limited

The spatial metrics are strong for a pilot, but the positive label set is still small and route clustered.

0.837

spatial AP

Primary ranked-retrieval metric for model selection.

0.971

ROC AUC

Separates labeled positives from negatives across thresholds.

100%

precision @10

Top-ranked validation targets remain highly precise.

16/17

field recovery

94.1% of known field points recovered within 15 m in the full layer.

Interpretation for researchers

The browser queue is intentionally decluttered for usability, so field-recovery analysis should use the full prediction layer. The next validation step is to collect more explicit no_culvert labels and compare models on the same route-aware split.

The next gains depend on data and protocol

Better model classes can help, but the largest improvements are likely to come from stronger labels, source layers, and validation design.

Bottleneck	What would help
Sparse negative labels	Add explicit no_culvert checks and label uncertainty during field review.
Prototype GIS layers	Replace Census roads and water with official NYSDOT, county, and drainage layers.
Hydrology features	Operationalize flow accumulation, drainage area, stream order, local slope, and catchment features.
Route-clustered reports	Use route-aware holdouts, active learning routes, and a shared error taxonomy.

The next phase is deeper validation

The prototype is ready for technical review and can be extended into a stronger research-grade culvert discovery workflow.

MODEL AND DATA ROADMAP

- **Hydraulic feature design**
Add drainage area, flow accumulation, local slope, stream order, catchment context, and embankment indicators.
- **Stronger field labels**
Collect more explicit no_culvert checks and route-balanced validation labels to reduce bias.
- **Model comparison**
Benchmark the current soft-voting baseline against alternative models on the same candidates and spatial split.

QUESTIONS FOR REVIEW

What I would value feedback on:

1. Which hydraulic predictors matter most?
2. Is the validation design defensible?
3. Which errors should be field-checked first?

Goal after review: identify the strongest path from working prototype to publishable research system.